To: Professor Davis

From: Walker Stone

Date: 3/13/2025

Subject: The Effect of Class Size on Student Achievement

I. Introduction

Schools are always looking for a way to optimize class size and test scores for students, in this policy memo will present evidence from a Regression Discontinuity Design (RDD), used compiled evidence from Israeli elementary schools' data. This memo will utilize that data to estimate the impact of class size on test scores. Using these findings, this memo will provide future policy recommendations to optimize class size.

II. Comparisons

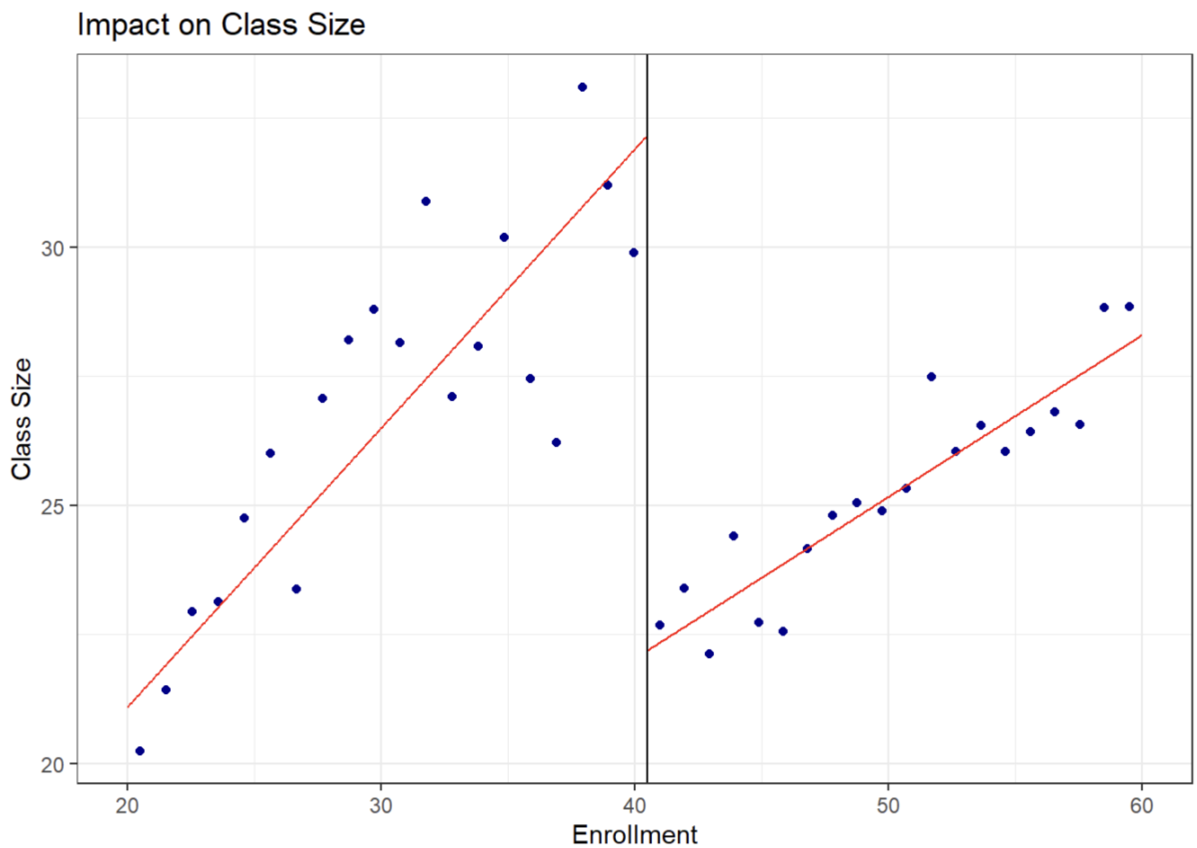
In the typical comparison would focus on the difference between large and small classes, however this is guaranteed to have biased estimates because it does not account for any other variables. The most common bias in that comparison would be selection bias, due to no randomization or common factors such as rich vs. poor schools and how much attention each student is given in class. That bias makes it impossible to isolate for the casual effect of class size on test scores. This is where the RDD comes in, this approach allows us to fix the bias issues by utilizing a policy cutoff, Mainmonides' rule, that determines if students are placed in small or large class, ideally, we would like to randomize it but not possible in this case. In Israel, Mainmonides' rule states that schools with 41 students must split the class into two, and no class can be bigger than 40 students. This creates a comparison between those classes right above 40 and below 40. Which means any differences in test scores can be directly shown in class sizes, because of the schools being so similar in every other aspect. In turn, creating an estimate of casual effect of smaller classes on test scores, without randomization.

III. Research

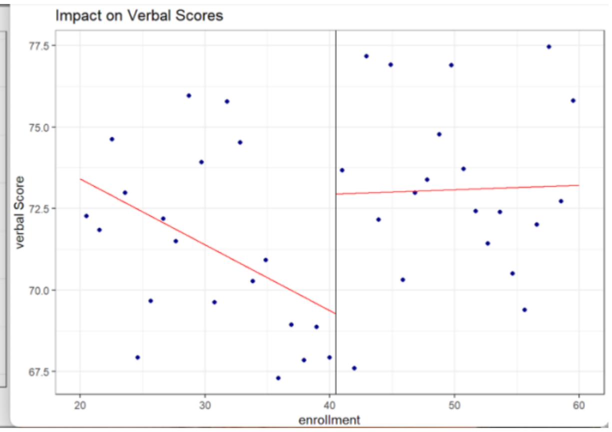
In this research, we observed data from some quasi-experimental studies, which estimate causal effects using differences in class size rules, providing a large portion of the data on class size impacts. Quasi-experiments, as opposed to randomized experiments, utilize external factors or policy norms that allow for comparisons. In Israeli schools, Angrist's Regression Discontinuity Design (RDD), which is based on Maimonides' Rule, is the most significant quasi-experimental study on this subject. According to their findings, "smaller classes significantly improve test scores, particularly in math" (Angrist & Lavy, p. 544). If everyone strictly followed Maimonides' Rule, we would expect class size to drop by 10 students, when schools surpass 40. The actual change in class size from Angrist & Lavy would be a reduction of 5 to 7 students. From this memo's regression we found that the data suggested a reduction of 9 students. The other well-known quasi-experimental study, Jepsen & Rivkin (2009) are another well-known quasi-experimental study that looks at California's statewide policy to reduce class sizes.

IV. Data

This memo used the dataset, grade5.csv, which has data from 5th grade test scores of Israeli public schools in 1991 (Angrist & Lavy). Variables in the dataset include class size (classize), class that are above 40-student threshold (above40), average math test scores (avgmath), average verbal test scores (avgverb), percentage of disadvantaged students (disadvantaged), fraction of female students (female), and an indicator for religious school status (religious). To represent, the impact of test scores of class size on tests scores, we created binned scatter plots. A binned scatter plot groups nearby observations into bins and plots their averages which reduces unneeded data while maintain trends in the data. Binned scatter plots allow us to show discontinuity at the 40-student enrollment threshold.



We first focused on class size as a function of enrollment, highlighting the expected drop in class size at the 40-student cutoff. This graph details the massive drop off once classes surpass 40 students, we also see similar effects in math and verbal scores.



In all these graphs, we see two sided graphs, where all the data on the left is from one single class, all the way up to 40 students. On the right side of graph, we see that these classes have been split into two and are producing higher schools due to the fact they have been split up and we are able to see the effect of class size. The discontinuity in these graphs shows the causal effect of class size on test scores. Math scores benefit more from class size reductions than verbal scores.

i. Regression Results

We estimated two regression models to show the impact of class size on student performance, the regression we used controlled for school characteristics, to isolate the causal effect of class size. Here are the results of test scores, of schools that were above the 40-student threshold that were required to split into two classes. This above40 variable, is significant because it allows us to capture the effect of small vs. large class sizes.

The effects of class size on math scores: above40 = 3.7202, p-value: 0.01

The effects of class size on verbal scores: above40 = 3.56904, p-value: 0.01

Change in Class size: above40 = -9.864, p-value: 0.01

There is significant evidence that reducing class size leads to improvement in test scores, math more than others.

V. Policy Recommendations

Policy recommendations to reduce classes could come in many forms, and each new policy could have positive or negative effects, that the data could never find. Jepsen & Rivkin (2009) detailed the effects of class reduction could have on other aspects than test scores, they inferred that teacher quality could decrease teacher quality, amongst the classes. So we would need a policy that would somehow alleviate that decrease while also increasing test scores. The main policy recommendation is to reduce class size by about 9 students in this data, this would increase test scores among math and verbal. A reduction from 40 to 35, a decrease of 5, would lead to 18.601 pp increase in math scores, (3.7202 x 5), while we would see a 17.845 pp increase in verbal scores. If we decided to decrease the class size anymore, we would likely not see the same type of increase in math and verbal test scores, a reduction of 20 to 15 would have very minimal marginal returns. Also, splitting up the classes that much would lead to teacher quality dilution, which Jepsen & Rivkin suggested. Overall, district should focus on strategically reductions, district to district, not a nationwide policy, because we would see teacher quality decrease and small marginal returns on test scores.

VI. Conclusion

This memo finds that class reductions can lead to increase in test scores, especially math.

Unfortunately, if reduction is too great, we could see a decrease in teacher quality, so for the most optimal reduction, we would need to constrain for teacher quality. Also, we see diminishing returns if reductions are too great. Overall, class reductions should be district by district decision, strategically constraining for teacher quality, would lead to the highest increase in student performance.

Works Cited

- Angrist, Joshua D., and Victor Lavy. "Using Maimonides' rule to estimate the effect of class size on scholastic achievement." The Quarterly Journal of Economics 114, no. 2 (1999): 533-575.
- Jepsen, Christopher, and Steven Rivkin. "Class size reduction and student achievement the potential tradeoff between teacher quality and class size." Journal of Human Resources 44, 1 (2009): 223-250.